

Business Plan and Final Requirements for Full Deployment of the Online Morabaraba Multiplayer Platform

ELEN3020A – Group 3

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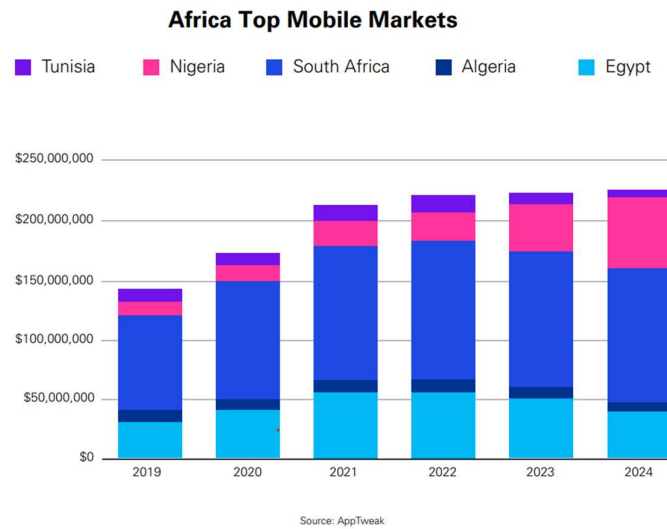
1. Executive summary

The online Morabaraba platform is a multiplayer game designed not only for entertainment, but to preserve and promote traditional African culture through digital technology. The platform allows players from different locations to interact in an online space while learning the rules, strategies and cultural significance of the game. Digitising Morabaraba helps to preserve traditional African games that could suffer in the current digital era. The platform demonstrates the ability to bridge the cultural gap between tradition and the modern day, and paves the way for other traditional games to also be adapted into a digital medium for future generations.

2. Market analysis

2.1 African digital gaming

Africa has a rapidly growing digital gaming market, especially in the mobile sector, with South Africa as its top earner. As of 2025, South Africa's game industry was projected to grow from \$266 million to \$333 million by 2026. Other top contributors include Egypt, Nigeria and Algeria [1].



2.2 Traditional games preservation

Traditional African games are seen as symbols of African identity, culture and resilience. Organisations such as the African Traditional Sports and Games Confederation (ATSGC) preserve the continent's heritage by bridging the gap between tradition and the modern world. The online Morabaraba platform contributes to the safeguarding of African games by digitally preserving the game, thus making it more attractive and accessible to younger generations. [2]

2.3 Online multiplayer market

The global online multiplayer games market is estimated to be worth \$95.75 million in 2026, with China being its main contributor. Online gaming platforms on cloud-based services facilitate 1.2 billion matchmaking sessions monthly, and serve 720 million unique users. The Middle East and Africa added over 95 million new multiplayer gamers in 2023, totalling 180 million players. Countries like UAE, Egypt and South Africa added a 28% growth in multiplayer platforms. [3]

2.4 Target market

- The main target groups of this platform are African gamers aged 12 and up, educators, and cultural conservation platforms.
- 50 000 registered users by year 1.
- English-speaking nations.

3. Financial aspects of the platform

3.1 Monetisation

In order to fund the maintenance and additional fees of this application, ads and monetary subscriptions will be implemented.

Therefore, there will be two versions available to the player depending on their subscription status:

Free-to-play version

- Players can:
 - create accounts and play all available traditional games;
 - play up to ten games a day;
 - access the matchmaking and ranked ladder system;

- participate in standard free-to-play tournaments, ran by the community;
- view basic game histories and player statistics; and
- interact in community chat and forums.

Paid subscription version:

- Players can:
 - play an unlimited number of games a day;
 - remain uninterrupted by ads anywhere on the platform;
 - view in-depth match statistics;
 - unlock all game themes, bar the purchasable themes;
 - access subscription-only tournaments; and
 - enjoy faster matchmaking and queue times.

Purchasable themes will be additional themes, available to all players for a set price per theme.

Themes will consist of board themes, game piece themes, avatars, emotes and profile backgrounds. Themes will not impact the gameplay.

3.2 Estimated cost for full project

For financial purposes, a scenario of 500 000 monthly active users, with 20 000 concurrent users during peak times will be considered for this cost breakdown. All values will be displayed in rands, using the current exchange rate as at 11 May 2026 of R16,39 to the US dollar.

- **Unity services**
 - **Unity multiplayer:**
Unity Relay is used for multiplayer networking, active game sessions, and matchmaking support. Relay pricing is approximately R2.62 per concurrent user-hour [4]. For an estimated 20 000 concurrent users during peak periods, the estimated total cost is approximately R52 447,99/month.
 - **Bandwidth:**
Bandwidth is used for gameplay data transfer, voice communication, video communication, matchmaking traffic, and backend service communication. Using a bandwidth cost of R2,29 per GB, estimated monthly infrastructure costs per user range from negligibly small for gameplay-only usage (R1,15-R2,79 with

voice chat) and up to approximately R32,78 per user per month when full video communication and backend services are included.

For an estimated 500 000 monthly active users, total bandwidth consumption is approximately 413 TB per month. Given Unity-style bandwidth pricing, this results in a significant infrastructure cost compared to other Unity services, with the total bandwidth cost amounting to R946 686,28/month.

- **Cloud storage:**

Unity Cloud Save is used to store structured user information such as profiles, rankings (ELO), match history, and settings. For an estimated 500 000 monthly active users, user data storage remains relatively small due to lightweight profile and match records. Estimated total storage is approximately 5-25 GB.

Given Unity-style storage pricing and operation costs, this results in a minimal infrastructure cost compared to bandwidth and computing. The total cloud storage cost will therefore be R8 195,00-R32 780,00/month.

- **Dev salaries**

In order to keep up with the maintenance needed to keep the full system platform running as intended, a minimum viable team size required would be 15 people. In South Africa, Unity developers are being paid an average of R250 000,00 per annum [5]. Therefore, total developer salary costing will be R312 500,00/month.

- **Marketing**

- Marketing for the platform would primarily focus on digital user acquisition through social media advertising, search engine marketing, influencer partnerships, and gaming community promotion using platforms such as Meta Ads [6] , Google Ads [7], and TikTok Ads [8].
- Industry research on mobile and online gaming platforms estimates average Cost Per Install (CPI) values of approximately R16,39-R57,36 depending on the target region and platform [9]. Initial large-scale user acquisition [10] could therefore cost approximately R16,3 million, with ongoing monthly marketing and retention campaigns costing in the region of R1,6 million per month.

- **QA testing**

- Testing will be conducted internally by the development team as well as through community beta testers. Developers will be responsible for functional testing, bug fixing, multiplayer synchronisation testing, and backend validation, while external beta testers will assist with gameplay feedback, usability testing, balance evaluation, and real-world network testing across different devices and connection conditions.

- Rather than a direct financial compensation, beta testers will be incentivised through early access to new features, mechanics, and platform updates before public release.

In Summary:

Category	Subcategory	Estimated Cost
Unity Services	Unity Relay (20 000 concurrent users)	R52 447,99/month
Unity Services	Bandwidth Infrastructure (413 TB/month)	R946 686,28/month
Unity Services	Cloud Storage	R8 195,00 – R32 780,00/month
Developer Salaries	15 Unity Developers	R312 500,00/month
Marketing	Ongoing Marketing & Retention Campaigns	~R1 600 000,00/month
Marketing	Initial User Acquisition Campaign	~R16 300 000,00 once-off
Total Monthly Operating Cost		~R2 919 829,27 – R2 944 414,27/month

4. Full proposed system

The full proposed system, taking into account all functionality as currently implemented in the MVP, as well as functionality to be implemented at a later stage, is as follows:

4.1 Core game system

- Morabaraba (Twelve Men's Morris) where the three-phase game is complete with all the rules: placing, moving, capturing, mill creation, flying, and win conditions.
- Nine Men's Morris variant where the game is scaled with the players each having nine pieces on the same board structure.
- Six Men's Morris variant where the game is more compact, with players having six pieces each on a smaller board.
- A rewind button where the previous two moves can be rewound, a function which each player can only use three times per game.
- Time controls where per-turn time limits can be set (forfeit on timeout) and overall game timers can be chosen by the host.

- Win/Loss and Draw detection where all end conditions are automatically detected: two pieces left, no legal move, or overall game timer runs out.

4.2 Multiplayer and networking

- Online 1v1 multiplayer where two players connect over the internet and play in real time via Unity Relay with lobby codes.
- Client-authoritative architecture where game logic runs on the client, a host authority controls turn state, and everything is synced via Unity Netcode for GameObjects.
- A lobby system where a relay-based lobby with sharable join codes is used to connect host and guest with no exposed IPs or dedicated servers required.
- Turn-based synchronisation where network variables and RPCs synchronise turn state, gameboard, and move validation across all connected clients.
- Spectator mode where additional players join as spectators with their inputs disabled while the game state is broadcast read-only to all spectators.
- A chat feature where players can chat with each other using in-game text chat.

4.3 AI opponents

- Single-player vs AI, where players can start a game against a computer opponent without needing a second human player online.
- Difficulty levels such as easy, medium and hard, where each represents a different depth of game tree search or heuristic quality.

4.4 User accounts, data and statistics

- Username and password authority where players register with a unique username and password, meeting security criteria using Unity Player Authentication.
- Persistent save data, where game state and results saved post-match will be stored and synced to Unity Cloud.
- Player statistics, where wins, losses and draws will be tracked via a statistics manager with the profile menu updating dynamically, and accessible from the main menu or in game menu.
- A match history database, where records of past matches are stored for players' personal review and replay.

- A ranking system where players are ranked locally and globally with leaderboards displayed in-app.

4.5 Tournaments and competitive play

- A tournament mode system, where there are brackets with automatic scheduling, result tracking and advancement.
- Prize and result tracking, where tournament results are recorded and prize distribution is tracked through platform administration tools.

4.6 Additional African games and cultural content

A number of additional digital African games and cultural content will be packaged with the final Morabaraba online multiplayer game. These include:

- *Mancala*: a strategic turn-based game played with stones or beans, where the goal is to capture opponents' pieces.
- *Dibeke*: A South African game that blends soccer and tag.
- *Intonga*: South African stick-fighting.
- *Jukskei*: South African traditional sport where you try to knock down a wooden peg placed in a sand pit with a wooden pin.
- Cultural content will include an in-app section that explains the history and significance of each game.

5. Implemented functionality and planned functionality

5.1 Implemented functionality

The following functionality has been achieved and implemented within the MVP to date:

- Core game system
 - Morabaraba game logic
 - Game variant of Six Men's Morris
 - Rewind function
 - Timer controls
 - Winner detection and announcement
 - Visual board and UI

- Multiplayer and networking
 - Online multiplayer 1v1
 - Client-authoritative architecture
 - Lobby system with join code
 - Turn-based synchronisation
- User accounts, data and statistics
 - Unity authentication
 - Player statistics
 - Persistent save data

5.2 Planned future functionality

The following is the functionality that is planned for implementation within the MVP in the future:

- Core game system
 - Game variant of Nine Men's Morris
- Multiplayer and networking
 - Spectator mode
 - Chat feature
- AI opponents
 - Single-player vs AI
 - Difficulty levels
- User accounts, data and statistics
 - Match history database
 - Ranking system
- Tournaments and competitive play
 - Tournament mode
- Additional African games and cultural content
 - Mancala variants and other African gameplay added
 - Cultural educational content

6. References

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